

Research on sequence over past centuries has obtained many interesting findings. A well-known but unsolved problem is the Collatz conjecture. Given any input x , if it is odd then multiply it by three, add it by one, and then divide it by two; otherwise, divide it by two. Applying the rule repeatedly will generate a sequence which ends at one. Write a program to generate such sequence for any input.

Input

The input starts with an integer m as the number of cases. Each case contains one integer, which denotes the value of x .

Output

For each case, output the sequence of x in the format:

“The sequence starts from x is: ...”

Sample Input

2
6
8

Sample Output

The sequence starts from 6 is: 6 3 5 8 4 2 1.
The sequence starts from 8 is: 8 4 2 1.